

Radio Physical Layer Specification

For the non-mode-0 steps $M + 1 \leq k \leq M + K$, and for each phase measurement period $1 \leq p \leq P[k]$, the average transmitted frequency $f[k, p]$ shall satisfy

$$|f[k, p] - f_E[k]| < 10\text{kHz}$$

for 95% of CS tone transmissions within non-mode-0 steps.

The center frequency of the transmitted CS_SYNC packet within any non-mode-0 step shall not deviate by more than 20 kHz with respect to the expected transmit frequency $f_E[k]$, for 95% of CS_SYNC transmissions within non-mode-0 steps.

The center frequency of the transmitted CS_SYNC packet within any mode-0 step $1 \leq k \leq M$ shall not deviate by more than 20 kHz with respect to the average frequency of the transmitted CS tone $f[k, 1]$, for 95% of CS_SYNC transmissions within mode-0 steps.

3.6 LE HDT PHY

[HDT-PHY_VSr01]

LE HDT PHY packets shall use a modulated signal at a 2 Msym/s rate with a symbol period of $T_s = 0.5 \mu\text{s}$. The symbol timing accuracy shall be less than ± 50 ppm. The modulation shall employ square-root raised cosine pulse shaping to generate the equivalent lowpass information-bearing signal $v(t)$. The output of the transmitter shall be a bandpass signal that can be represented as

$$S(t) = \text{Re}\left(v(t)e^{j2\pi F_c t}\right),$$

with F_c denoting the carrier frequency. $v(t)$ shall be generated according to

$$v(t) = \sum_{k=0}^{N \div \log_2(M) - 1} S_k p(t - kT_s)$$

The pulse shaping function $p(t)$ and the complex-valued information symbols S_k in a symbol stream are defined in Section 7. N is the bitstream length and M is the number of constellation points as specified in Section 7.3.

Note: The pulse shaping function $p(t)$ is the same as the one in [Vol 2] Part A, Section 3.2.1.3. [ID28319_VSr03]

The accuracy of the transmitted symbols S_k shall be measured in terms of the root mean square (RMS) error vector magnitude (EVM) as detailed in Appendix C over 1000 symbols. The RMS EVM shall be measured over 1500 packets. Table 3.6 shows the maximum transmitter RMS EVM per data rate. The Control Header RMS EVM and the payload RMS EVM for each packet shall be lower than the maximum transmitter EVM defined in Table 3.6.



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LE HDT PHY Rate	Control Header RMS EVM Limit	PDU Header and payload RMS EVM Limit
HDT2	-10 dB	-10 dB
HDT3		-13 dB
HDT4		-16 dB
HDT6		-19 dB
HDT7.5		-22 dB

Table 3.6: EVM per data rate

3.6.1 Pre-packet emissions

[ID28876_VSr03]

The duration from the first instant the RF output power rises above a level 35 dB below that of the average packet output power to the first instant where it reaches a level 1 dB below the average packet output power shall not be more than 4 μ s.



4 RECEIVER CHARACTERISTICS

The reference sensitivity level referred to in this section is -70 dBm for GFSK PHYs. The reference sensitivity levels for the LE HDT PHY are shown in Table 4.5. The packet error rate corresponding to the defined bit error ratio (BER) for the GFSK PHYs or the block error rate (BLER) for the LE HDT PHY shall be used in all receiver characteristic measurements. A block error is considered to be a data block containing one or more errors after being processed by the receiver as specified in [Vol 6] Part B, Section 3.4.

[HDT-PHY_VSr01]

4.1 Actual sensitivity level

The actual sensitivity level is defined as the receiver input level for which the BER specified in Table 4.1 is achieved.

Maximum Supported Payload Length (bytes)	BER (%)
1 to 37	0.1
38 to 63	0.064
64 to 127	0.034
128 to 255	0.017

Table 4.1: Actual sensitivity BER by maximum payload length

The actual sensitivity level of the receiver for a given PHY shall be as specified in Table 4.2. This shall apply when receiving signals specified in Section 3 together with any combination of the following allowed parameter variations:

- Initial frequency offset
- Frequency drift
- Symbol rate
- Frequency deviation

PHY	Sensitivity (dBm)
LE Uncoded PHYs	≤ -70
LE Coded PHY with S=2 coding	≤ -75
LE Coded PHY with S=8 coding	≤ -82

Table 4.2: Receiver sensitivity for a given PHY

[HDT-PHY_VSr01]



7 LE HDT PHY SYMBOL STREAM PROCESSING

[HDT-PHY_VSr01]
[ID27963_VSr02]

The LE HDT PHY uses training sequences and uses three coherent modulation schemes as well as convolutional coding and puncturing to provide five data bit rates (see Section 7.3). From the training sequences, only the Terminating symbols are generated using coherent modulation, all other training sequences are not modulated data bits (see Section 7.4). The symbol rate for all symbols in a packet is specified in Section 3.6. The data rate used is indicated in each packet, which allows the rate to be changed on a packet-by-packet basis. A summary of the LE HDT PHY for the PDU Header and Payload symbols for packet format 0, and for the Payload Zone symbols for packet format 1, is shown in Table 7.1, where the effective bit rate is the coded data rate of the payload zone. [ID27963_VSr02]

LE HDT PHY Rate Name	HDT2	HDT3	HDT4	HDT6	HDT7.5
Effective Bit Rate	2 Mb/s	3 Mb/s	4 Mb/s	6 Mb/s	7.5 Mb/s
Modulation Scheme	$\pi/4$ QPSK	$\pi/4$ QPSK	8PSK	16QAM	16QAM

Table 7.1: Summary of the LE HDT PHY [ID27963_VSr02]

7.1 Packet symbols

[HDT-PHY_VSr01]

The general structure of LE HDT PHY packets is shown in Figure 7.1. There are three packet formats: short packet format, packet format 0, and packet format 1. [ID27963_VSr02]

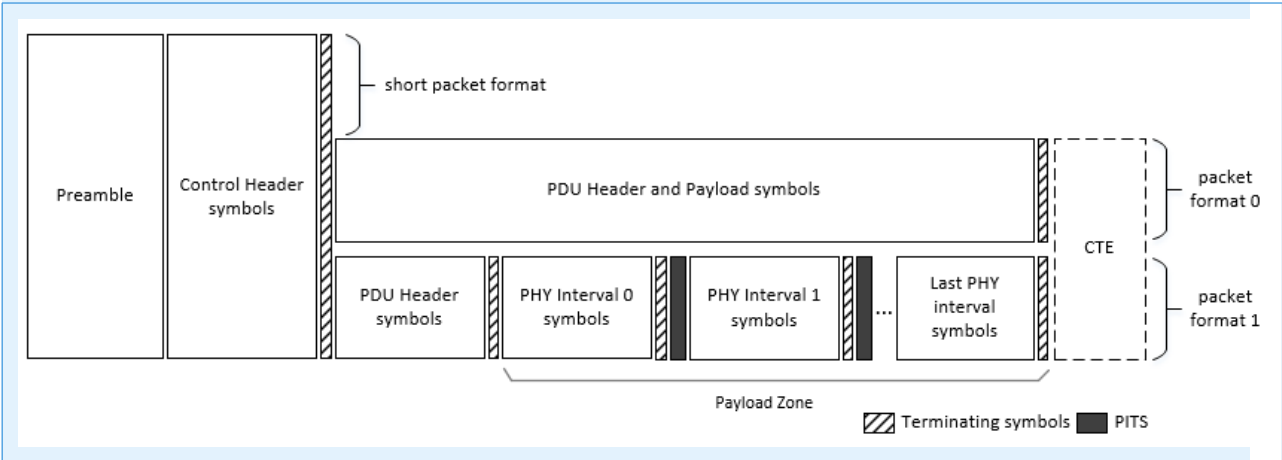


Figure 7.1: LE HDT PHY packet formats [ID27963_VSr02]



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All formats have a Preamble and a Control Header. Packet format 0 has the PDU Header and an optional payload transmitted in a single symbol stream. Packet format 1 has a separate PDU Header symbol stream followed by the optional Payload Zone, which contains the payload-carrying PHY Interval symbols of the packet (see [Vol 6] Part B, Section 2.7). Packet format 1 may also contain one or more PHY Interval Training Sequences (PITS) as specified in Section 7.4.5. Both packet format 0 and packet format 1 may include a CTE (see [Vol 6] Part B, Section 2.5). **[HDT-PACK_VSr03]**

7.2 PHY Intervals

[HDT-PHY_VSr01]

The Payload Zone in packet format 1 shall be split into intervals of a fixed number of symbols, each called a PHY Interval symbol stream. A terminating symbol sequence of two symbols (see Section 7.4.4) shall be transmitted after each PHY Interval. A PITS (see Section 7.4.5) shall be transmitted between a terminating symbol sequence of one PHY Interval and the start of the following PHY Interval symbol stream. The number of Payload symbols per PHY Interval shall be specified by the PHY INT field of the Control Header (see [Vol 6] Part B, Section 2.7.1) and shown in Table 7.2. **[HDT-PACK_VSr03]**

PHY INT	Payload Symbols per PHY Interval
0b00	128
0b01	256
0b10	384
0b11	512

Table 7.2: Number of PHY Interval Payload symbols

The PHY Interval symbol stream, plus a FEC termination bit sequence (see [Vol 6] Part B, Section 3.4.3) and any zero padding bits needed, is mapped into a sequence of modulated symbols. The PHY Interval bitstream shall be mapped into symbols as specified in Section 7.3. The total PHY Interval symbol stream length may vary depending on the data rate and is specified in Table 7.3. The last PHY Interval in a packet may be shorter if the symbol stream is shorter than the lengths specified in Table 7.2.

PHY INT	HDT2	HDT3	HDT4	HDT6	HDT7.5
0b00	133	132	131	130	130
0b01	261	260	259	258	258
0b10	389	388	387	386	386
0b11	517	516	515	514	514

Table 7.3: PHY Interval symbol stream length in number of symbols when including the payload-carrying symbols, FEC termination sequence, and zero padding



7.3 Modulation for the LE HDT PHY

[HDT-PHY_VSr01]

Each modulation scheme shall map each processed bitstream $\{b_n\}$, $n = 0, 1, 2, 3, \dots, N-1$ (see [Vol 6] Part B, Section 3.4) onto a corresponding symbol stream $\{S_k\}$, $k = 0, 1, 2, \dots, N/\log_2(M) - 1$ of complex-valued signal points (symbols), where $M=4$ for $\pi/4$ QPSK, $M=8$ for 8PSK, and $M=16$ for 16QAM. If the length of the binary bitstream N is not an integer multiple of $\log_2(M)$, then the last symbol of the sequence $\{S_k\}$ shall be formed by appending zero padding bits for symbol completion. The bit index n starts at 0 at the first transmitted bit. The bit index n is reset to $n=0$ after each terminating symbol sequence.

7.3.1 $\pi/4$ QPSK symbol mapping

[HDT-PHY_VSr01]

In $\pi/4$ QPSK, two bits are mapped to one symbol. Symbol S_k where index k is odd uses a phase mapping with a phase offset of $\pi/4$ relative to the symbols where k is even. In packet format 0, the symbol index k is reset to $k=0$ at the first PDU Header and Payload symbol. In packet format 1, the symbol index k is reset to $k=0$ at the first symbol of the PDU Header and at the first symbol of each PHY Interval. Table 7.4 and Figure 7.2 specify the $\pi/4$ QPSK symbol mappings. For example, if the first 8 bits of the processed bitstream are 0, 0, 1, 0, 1, 1, 0, and 0, then the first four $\pi/4$ QPSK symbols are $e^{j\pi/4}$, 1, $e^{-j3\pi/4}$, and j .

Bits ($n=4k+0, n=4k+1$)	Symbol S_{2k}	Bits ($n=4k+2, n=4k+3$)	Symbol S_{2k+1}
(0, 0)	$e^{j\pi/4}$	(0, 0)	$e^{j\pi/2}$
(0, 1)	$e^{j3\pi/4}$	(0, 1)	$e^{j\pi}$
(1, 0)	$e^{-j\pi/4}$	(1, 0)	e^0
(1, 1)	$e^{-j3\pi/4}$	(1, 1)	$e^{-j\pi/2}$

Table 7.4: $\pi/4$ QPSK phase mapping

Note: The symbols $e^{j\pi/2}$, $e^{j\pi}$, e^0 , and $e^{-j\pi/2}$ can be simplified to j , -1 , 1 , and $-j$, respectively. [ID28319_VSr03]



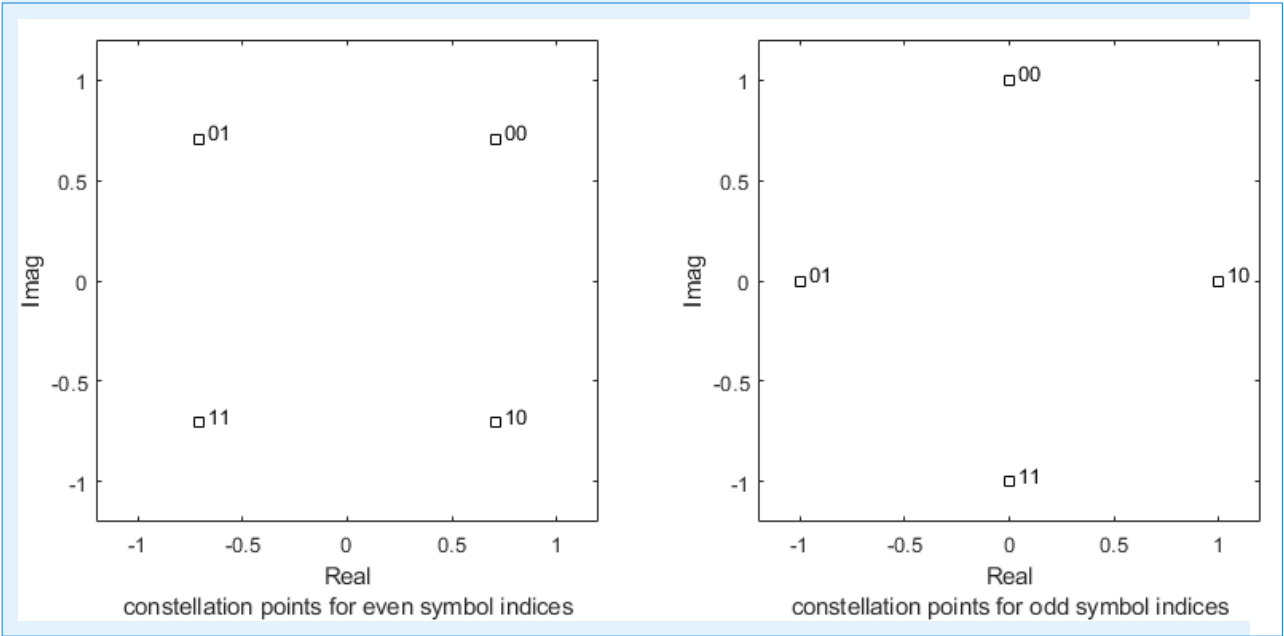


Figure 7.2: $\pi/4$ QPSK symbol mapping

7.3.2 8PSK symbol mapping

[HDT-PHY_VSr01]

In 8PSK, three bits are mapped to one symbol. Table 7.5 and Figure 7.3 specify the symbol mappings used in 8PSK. For example, if the first 6 bits of the processed bitstream are 0, 0, 1, 0, 1, and 1, then the first two 8PSK symbols are $e^{j\pi/4}$ and $e^{j\pi/2}$.

Bits (n=3k+0, n=3k+1, n=3k+2)	Symbol S_k
(0, 0, 0)	e^0
(0, 0, 1)	$e^{j\pi/4}$
(0, 1, 0)	$e^{j3\pi/4}$
(0, 1, 1)	$e^{j\pi/2}$
(1, 0, 0)	$e^{-j\pi/4}$
(1, 0, 1)	$e^{-j\pi/2}$
(1, 1, 0)	$e^{-j\pi}$
(1, 1, 1)	$e^{-j3\pi/4}$

Table 7.5: 8PSK phase mapping



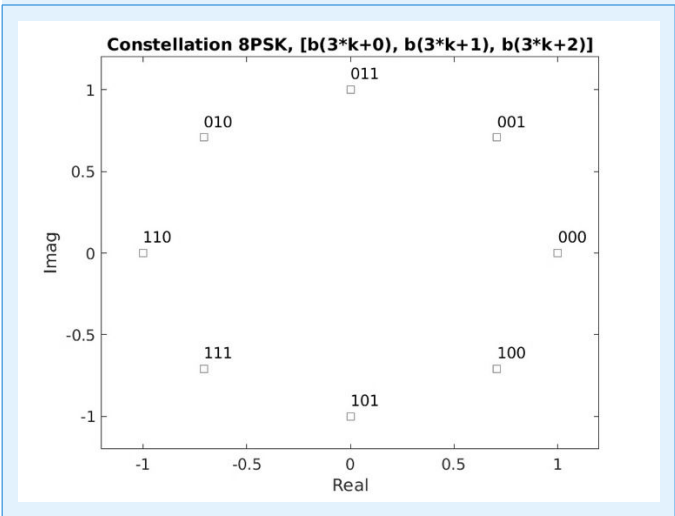


Figure 7.3: 8PSK symbol mapping

7.3.3 16QAM symbol mapping

[HDT-PHY_VSr01]

In 16QAM, four bits are mapped to one symbol. Table 7.6 and Figure 7.4 specify the symbol mappings used in 16QAM. The symbols in Table 7.6 shall be normalized by dividing the number in the right column by $\sqrt{10}$ to find the transmitted constellation points shown in Figure 7.4. For example, if the first 8 bits of the processed bitstream are 0, 0, 1, 0, 1, 1, 0, and 0, then the first two 16QAM symbols are $(-3 + 3j) \div \sqrt{10}$ and $(1 - 3j) \div \sqrt{10}$.

Bits (n=4k+0, n=4k+1, n=4k+2, n=4k+3)	Symbol $S_k \times \sqrt{10}$
(0, 0, 0, 0)	-3 - 3j
(0, 0, 0, 1)	-3 - 1j
(0, 0, 1, 0)	-3 + 3j
(0, 0, 1, 1)	-3 + 1j
(0, 1, 0, 0)	-1 - 3j
(0, 1, 0, 1)	-1 - 1j
(0, 1, 1, 0)	-1 + 3j
(0, 1, 1, 1)	-1 + 1j
(1, 0, 0, 0)	3 - 3j
(1, 0, 0, 1)	3 - 1j
(1, 0, 1, 0)	3 + 3j
(1, 0, 1, 1)	3 + 1j
(1, 1, 0, 0)	1 - 3j



Bits (n=4k+0, n=4k+1, n=4k+2, n=4k+3)	Symbol $S_k \times \sqrt{10}$
(1, 1, 0, 1)	$1 - 1j$
(1, 1, 1, 0)	$1 + 3j$
(1, 1, 1, 1)	$1 + 1j$

Table 7.6: 16QAM symbol mapping before normalization

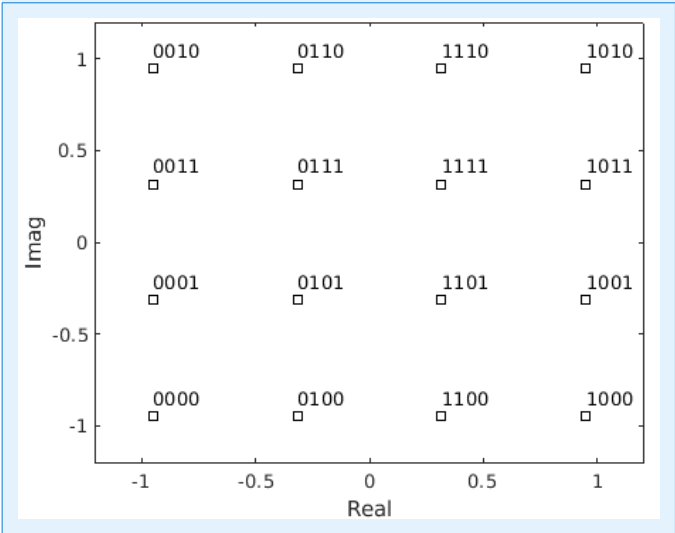


Figure 7.4: 16QAM symbol mapping

7.4 Training sequences

[HDT-PHY_VSr01]

LE HDT PHY packets carry multiple training sequences. All packets have a preamble, which consists of nine repetitions of the 2 μs short training sequence (STS) described in Section 7.4.1, the 2 μs guard interval (GI) described in Section 7.4.3, and finally two repetitions of the 8.5 μs long training sequence (LTS) described in Section 7.4.2. Figure 7.5 shows the preamble structure. [ID27963_VSr02] [ID28478_VSr02]

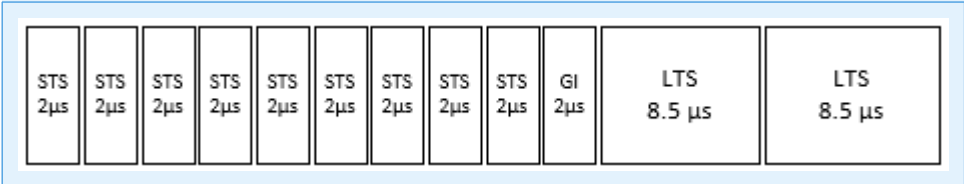


Figure 7.5: LE HDT PHY packet preamble structure [ID27963_VSr02]

Terminating symbols (see Section 7.4.4) are appended to each symbol stream and packet format 1 inserts a PHY Interval Training Sequence (see Section 7.4.5) between PHY intervals. [ID28478_VSr02]

7.4.1 Short Training Sequences

[HDT-PHY_VSr01]

A Short Training Sequence consists of the four symbols -1, -j, j, and 1 in transmission order.

7.4.2 Long Training Sequences

[HDT-PHY_VSr01]

A Long Training Sequence consists of a Zadoff-Chu sequence comprising 17 symbols, $x_u(0)$ to $x_u(16)$, in transmission order, where

$$x_u(k) = e^{-j\frac{\pi uk(k+1)}{17}} \times e^{j2\pi\frac{p}{17}}$$

The choice of u is specified in [Vol 6] Part B, Section 2.7.1.1. The value of p shall be derived from u as specified in Table 7.7. [HDT-PACK_VSr03]

u	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
p	2	7	12	12	4	13	8	14	15	13	12	10	5	15	14	0

Table 7.7: Mapping from u to p in the choice of LTS

7.4.3 Guard interval

[HDT-PHY_VSr01]

The first LTS field is preceded by the GI field, which is a copy of the last four LTS symbols (i.e., $x_u(13)$, $x_u(14)$, $x_u(15)$, and $x_u(16)$ in transmission order).

7.4.4 Terminating symbols

[HDT-PHY_VSr01]

Two terminating symbols are inserted after the Control Header symbols of each packet. In packet format 0, two terminating symbols are inserted after the PDU Header and Payload symbols. In packet format 1, two terminating symbols are inserted after the PDU Header symbols and after each PHY Interval symbol stream. The terminating symbols consist of symbols representing all zero bits using the modulation of the symbol stream being terminated. [ID28478_VSr02]

7.4.5 PHY Interval Training Sequence

[HDT-PHY_VSr01]

In packet format 1, the Payload is divided into intervals of fixed numbers of symbols. The length of an interval is specified by the PHY INT field of the Control Header (see [Vol 6] Part B, Section 2.7.1); the first PHY Interval starts with the first symbol of the



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Payload. The PITS shall be inserted between the end of the terminating symbols of one PHY interval and the start of the next PHY Interval. The PITS shall not be appended after the terminating symbols following the last PHY Interval. **[HDT-PACK_VSr03]**

The PITS consists of the six symbols -1, +1, +1, +1, -1, and +1 in transmission order.

Note: The symbols of the PITS do not represent modulated data bits. Therefore, they are not valid 16QAM symbols and are not all valid $\pi/4$ QPSK symbols.

7.5 Pulse shaping

[HDT-PHY_VSr01]

The symbol pulse shaping is achieved by applying a square root-raised cosine filter. The frequency response is defined as

$$|P(f)| = \begin{cases} 1 & 0 \leq |f| \leq \frac{1-\beta}{2T}, \\ \sqrt{\frac{1}{2} \left(1 - \sin\left(\frac{\pi(2|f|T-1)}{2\beta}\right) \right)} & \frac{1-\beta}{2T} \leq |f| \leq \frac{1+\beta}{2T}, \\ 0 & \text{elsewhere,} \end{cases}$$

where f is frequency in Hz, the roll-off factor β is set to 0.4, and T is the symbol period. T is set to 0.5 μ s. The time domain pulse-shaping filter $p(t)$ is the Inverse Fourier Transform of $P(f)$.



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- The Reason field shall be set to inform the Synchronized Receiver(s) why the BIG is about to be terminated. See [Vol 1] Part F, Controller Error Codes for a list of error codes and descriptions.
- The Instant field shall be set to the value of *bigEventCounter*₁₅₋₀ (see Section 4.4.6.3) when the BIG will be terminated.

2.7 Packet format for the LE HDT PHY

[HDT-PACK_VSr03]

The packet format described in this section is defined for the LE HDT PHY and is used for packets on all physical channels that support the LE HDT PHY.

This format contains up to four layers (e.g., the PDU Header field of a packet can contain a Tx field, which can contain LL Descriptor fields, which each contain an LLID field). To make the layering more obvious, the fields at the highest layer are sometimes referred to as zones and the fields at the next layer down as portions. The bottom two layers, when the structure is that deep, are just called fields.

The overall packet format is shown in Figure 2.93. Each packet consists of two mandatory zones and three optional zones. The mandatory zones are the Preamble and Control Header. The optional zones are the PDU Header Zone, Payload Zone, and CTE.

LE HDT PHY Packet				
Preamble (37 μs)	Control Header (32 μs)	PDU Header Zone	Payload Zone	CTE

Figure 2.93: Link Layer packet format for the LE HDT PHY

The packet format has three variants called short format, packet format 0, and packet format 1. The format used is indicated in the Control Header. Packets using the short format shall not have a PDU Header Zone, Payload Zone, or CTE. Packets using packet format 0 or packet format 1 shall have a PDU Header Zone and may have a Payload Zone, CTE, or both. Figure 2.94 shows the major differences between the three variants.



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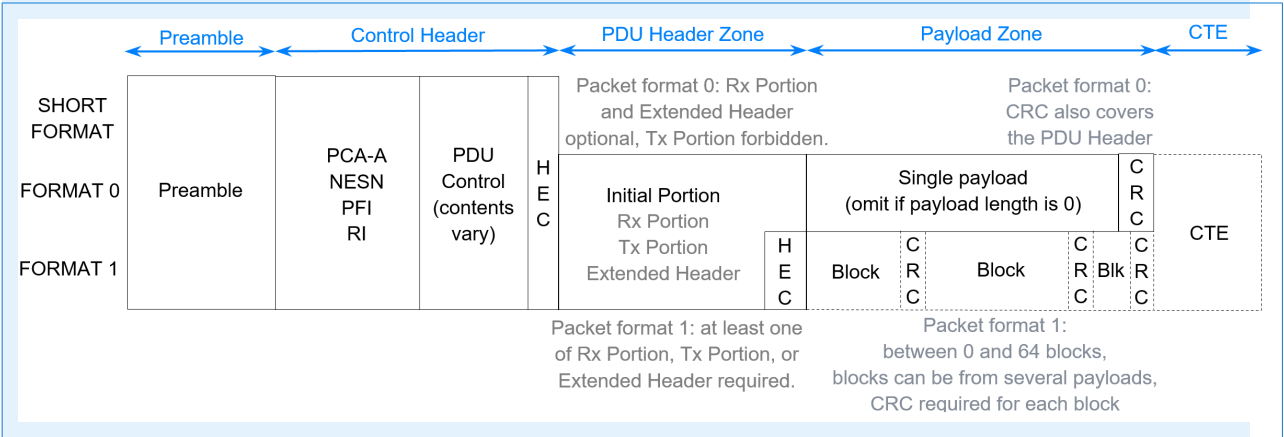


Figure 2.94: Packet format variants

Figure 2.95 shows the parts of Figure 2.94 that make up a PDU. In the case of packet format 1, this example shows blocks from two different payloads and so there are two separate PDUs, each with their own LL descriptor and blocks.

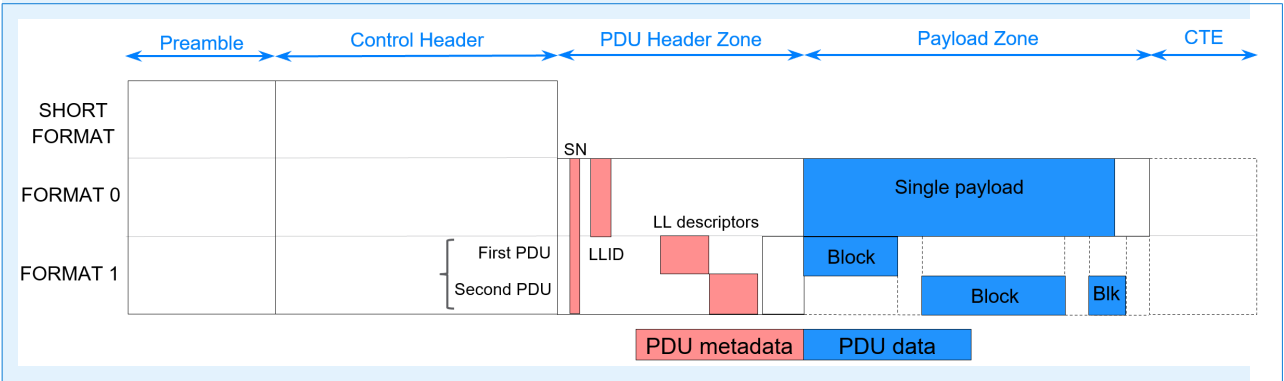


Figure 2.95: PDU components within a packet

A packet in short format cannot contain any data for transmission. It can only acknowledge complete payloads and only from the start of the payload window by using NESN (see Section 4.5.9.2.2). The only other fields it can contain are MD, CIE, and SNR (see Section 4.5.6, Section 4.5.13.4, and Section 2.7.2.4.1 respectively).

A packet in packet format 0 shall contain a single PDU. A packet in packet format 1 may hold up to four separate PDUs. The header information of the PDU(s) is stored in the PDU Header Zone of the packet. The payloads and MICs (if included) of the PDU(s) are stored in the Payload Zone.

The Preamble is described in [Vol 6] Part A, Section 7.4. The Control Header is described in Section 2.7.1. The PDU Header Zone is described in Section 2.7.2. The Payload Zone is described in Section 2.7.3. The CTE is described in Section 2.5.



2.7.1 Control Header

[HDT-PACK_VSr03]

The Control Header has the format shown in Figure 2.96:

Control Header						
PCA-A (16 bits)	NESN (3 bits)	PFI (1 bit)	RI (3 bits)	RFU (1 bit)	PDU Control (9 bits)	HEC-C (24 bits)

Figure 2.96: Control Header format

The PCA-A field shall be set to bits 39:24 of the Physical Channel Address (see Section 2.7.1.1).

The NESN field is described in Section 4.5.9.2.2 for ACL and CIS packets and is RFU for BIS packets.

The PFI field indicates the format used for the packet. The PFI field shall be 0 for the short format and packet format 0, and shall be 1 for packet format 1.

The RI field indicates the data rate to be used for the PDU Header Zone and Payload Zone and shall be set as specified in Table 2.54.

RI Field	Data Rate
0b000	Short format: no PDU Header Zone or Payload Zone
0b001	2 Mb/s
0b010	3 Mb/s
0b011	4 Mb/s
0b100	6 Mb/s
0b101	7.5 Mb/s
0b110 0b111	Reserved for future use

Table 2.54: RI field values

Note: In packet format 1, the data rate of the PDU Header Zone can be less than the rate indicated in Table 2.54 because of the lack of puncturing (see [Vol 6] Part A, Section 3.6).

The PDU Control field is defined in Section 2.7.1.2.

The HEC-C field shall contain a CRC of all the other fields of the Control Header and is generated as specified in Section 3.1.1.

*Link Layer Specification***2.7.1.1 Physical Channel Address****[HDT-PACK_VSr03]**

The Physical Channel Address (PCA) is a 40-bit value. Each time the Link Layer needs a new PCA (except for a Broadcast Isochronous Stream (BIS)), it shall generate a new random value according to the requirements for random generation defined in [Vol 2] Part H, Section 2.

Each PCA shall meet the following requirements:

- It shall not be the PCA for any existing ACL connection or CIS on this device.
- It shall not be the PCA for any existing BIS or BIG Control logical link on this device.
- It shall not have the value 0x8E_8942_BED6.

It is intended that each Link Layer connection between any two devices and each BIS on the LE HDT PHY has a different PCA.

The Link Layer in the Central Role in the Connected State shall generate a new PCA when switching to the LE HDT PHY for the first time on an ACL connection. The PCA is sent in the LL_PHY_UPDATE_IND PDU. If the connection is moved to a GFSK PHY and then back to the LE HDT PHY, then it shall continue to use the same PCA; that PCA shall remain associated with the connection in the interim.

The Link Layer in the Central Role in the Connected State shall generate a new PCA for each CIS it creates on the LE HDT PHY. The PCA is sent in the LL_CIS_IND PDU.

The Link Layer in the Isochronous Broadcasting State shall generate a new Seed PCA (SPCA) for each BIG it creates on the LE HDT PHY. The SPCA shall be a random 40-bit number that meets the following requirements:

$$\begin{aligned} \text{SPCA}_{23-16} &= 0xFF - \text{SPCA}_{15-8} \\ \text{SPCA}_{27} &= \text{SPCA}_{15} \\ \text{SPCA}_{30} &= \text{SPCA}_{24} \neq \text{SPCA}_{15} \\ \text{SPCA}_{33} &= 0 \\ \text{SPCA}_{31} &= 1 \end{aligned}$$

For any pair of BIGs transmitted by the same device, the SPCA₁₅₋₀ values shall differ in at least two bits.

The PCAs for the constituent BIS(es) and the BIG Control logical link (see Section 4.4.6.7) are derived from the SPCA.

For each BIS logical transport, the PCA shall be equal to the SPCA XORed with a diversifier word (DW) for that logical transport derived from a Diversifier (D) as follows:



$D = ((35 \times n) + 42) \bmod 128$ where n is the BIS_Number, or 0 for the BIG Control logical link

$DW = 0bD_0D_0D_0D_0D_0D_0D_1D_6_D_10D_5D_40D_3D_20_00000000_00000000_00000000$

For example, if $n=1$, $D=77=0b01001101$ and $DW = 0xFD06_000000$.

If the random number and, in the case of an SPCA, the derived PCAs for the BISes and the BIG Control logical link do not meet the above requirements, then new random numbers shall be generated until the requirements are met.

Each PCA has an associated value u between 1 and 16 (see [Vol 6] Part A, Section 7.4.2) such that $u = PCA_{31:0} \bmod 15 + 1$. The value $u = 16$ is reserved for future use.

2.7.1.2 PDU Control field

[HDT-PACK_VSr03]

In the short packet format, the PDU Control field shall have the format shown in Figure 2.97.

PDU Control field		
MD/CIE (1 bit)	RFU (3 bits)	SNR (5 bits)

Figure 2.97: PDU Control field in the short packet format

In ACL packets, the MD/CIE field shall be the MD field defined in Section 4.5.6.

In CIS packets, the MD/CIE field shall be the CIE field defined in Section 4.5.13.4.

In BIS packets, the MD/CIE field shall be reserved for future use.

The SNR field is defined in Section 2.7.2.4.1.

In packet format 0, the PDU Control field shall indicate the total length of the PDU Header Zone and Payload Zone, excluding the CRC, in octets. The length is 9 bits.

In packet format 1, the PDU Control field shall have the format shown in Figure 2.98.

PDU Control field		
PDU Header Zone length (6 bits)	PHY INT (2 bits)	RFU (1 bit)

Figure 2.98: PDU Control field in packet format 1

The PDU Header Zone length field shall indicate the total length of the PDU Header Zone, excluding the HEC-P Portion, in octets.

The PHY INT field is defined in [Vol 6] Part A, Section 7.2.

2.7.2 PDU Header Zone

[HDT-PACK_VSr03]

This section applies to packet formats 0 and 1 but not to the short format.

The PDU Header Zone consists of five portions as shown in Figure 2.99.

PDU Header Zone				
Initial Portion (8 bits)	Rx Portion (ACL and CIS only)	Tx Portion (format 1 only)	Extended Header	HEC-P (format 1 only)

Figure 2.99: PDU Header Zone portions

The Initial Portion shall be present. The HEC-P Portion shall be omitted in packet format 0 and present in packet format 1. The presence of each of the Rx Portion, the Tx Portion, and the Extended Header is signalled by a bit in the Initial Portion as specified in Section 2.7.2.1. In packet format 1, at least one of the Rx Portion, Tx Portion, or Extended Header shall be present.

The PDU Header Zone shall not be more than 63 octets long, excluding the HEC-P in packet format 1.

2.7.2.1 Initial Portion

[HDT-PACK_VSr03]

The Initial Portion shall be a single octet with the format shown in Figure 2.100 in ACL packets, the format shown in Figure 2.101 in CIS packets, and the format shown in Figure 2.102 in BIS packets.

Bit	0	1	2	3-5	6	7
Format 0	XHP	RxPP	MD	SN	LLID	
Format 1					TxPP	RFU

Figure 2.100: Initial Portion for ACL packets

Link Layer Specification

Bit	0	1	2	3-5	6	7
Format 0	XHP	RxPP	CIE	SN	LLID	
Format 1					TxPP	RFU

Figure 2.101: Initial Portion for CIS packets

Bit	0	1	2	3-5	6	7
Format 0	XHP	0	CSTF	CSSN	LLID	
Format 1					TxPP	RFU

Figure 2.102: Initial Portion for BIS packets

The XHP field shall be 1 if the Extended Header is present in the PDU Header Zone and 0 otherwise.

The RxPP field shall be 1 if the Rx Portion is present in the PDU Header Zone and 0 otherwise. The Rx Portion shall not be present in BIS packets.

The TxPP field shall be 1 if the Tx Portion is present in the PDU Header Zone and 0 otherwise. The Tx Portion shall not be present in packet format 0.

The MD field is described in Section 4.5.6.

The CIE field is described in Section 4.5.13.4.

The CSTF and CSSN fields are described in Section 4.4.6.7.

The SN field is described in Section 4.5.9.2.2.

The LLID field is described in Section 2.4 for Data Physical Channel PDUs, Section 2.6.1 for CIS PDUs, and Section 2.6.2 for BIS PDUs.

2.7.2.2 Rx Portion

[HDT-PACK_VSr03]

The Rx Portion, when it is present, indicates whether the device is acknowledging the receipt of data sent by the peer device. The Rx Portion shall have the format shown in Figure 2.103.



Link Layer Specification

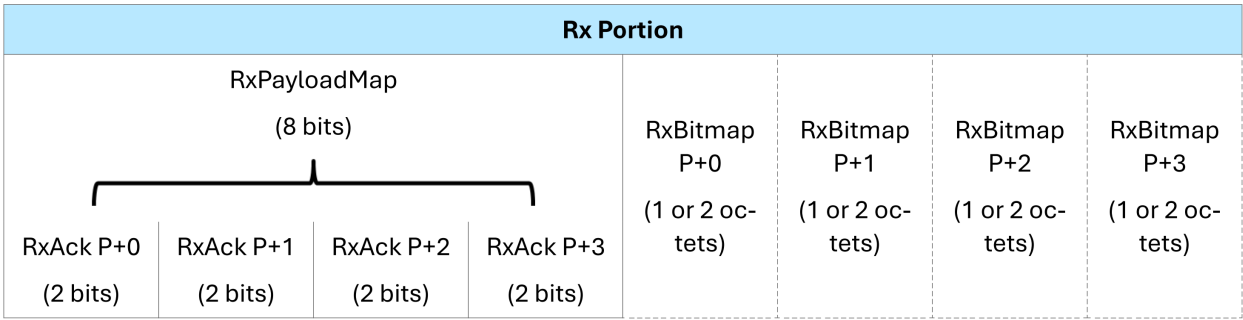


Figure 2.103: Rx Portion contents

Each RxAck P+k and RxBitmap P+k field corresponds to the payload with payload number *ExpectedPayloadNumber*+k (see Section 4.5.9.2.2). The RxAck P+k values and their effect on RxBitmap P+k are shown in Table 2.55.

RxAck P+k Value	RxBitmap P+k Size	Meaning
0b00	Omitted	Payload unacknowledged
0b01	Omitted	Payload acknowledged
0b10	1 octet	RxBitmap P+k specifies the status of each block in the payload
0b11	2 octets	

Table 2.55: RxAck values

The RxPayloadMap field should not have the value 0x00. If no payload or block needs its status updated, the Rx Portion should be omitted.

The RxAck P+0 field shall not have the value 0b01; NESN is increased instead (see Section 4.5.9.2.4).

If the RxBitmap field has only one octet, the omitted bits 8 to 15 are treated as being set. For example, a single-octet RxBitmap field with the value 0x42 has the same meaning as a two-octet RxBitmap field with the value 0xFF42.

The RxAck and RxBitmap fields are described further in Section 4.5.9.2.6.

A RxBitmap field should not have the value 0xFF or 0xFFFF (depending on RxAck); RxAck should be set to 0b01 instead (which can then cause NESN to be increased).

2.7.2.3 Tx Portion

[HDT-PACK_VSr03]

The Tx Portion, when it is present, indicates which payload blocks the device is sending. The Tx Portion shall have the format shown in Figure 2.104.



Link Layer Specification

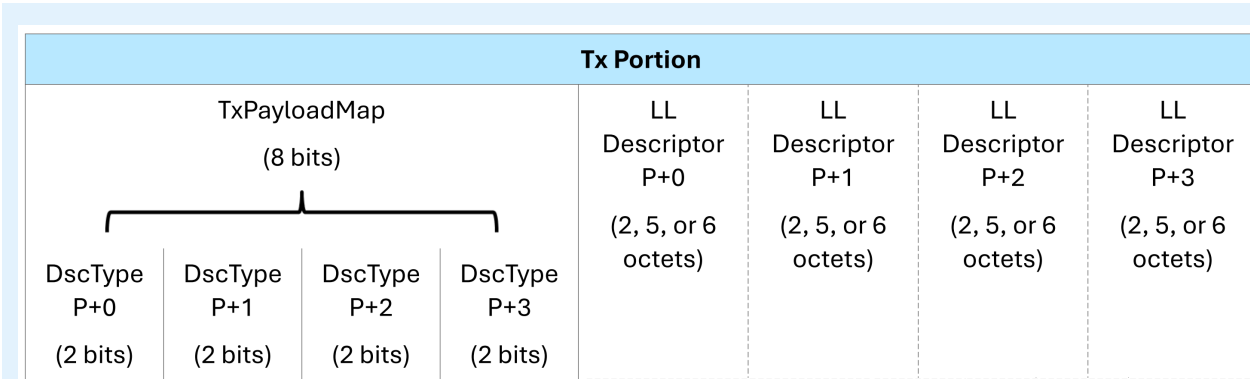


Figure 2.104: Tx Portion contents

Each DscType P+k field and LL Descriptor P+k corresponds to the payload with payload number *CurrentPayloadNumber*+k (see Section 4.5.9.2.2). If the Payload Zone contains one or more blocks of payload P+k (see Section 2.7.4), then LL Descriptor P+k shall be present; otherwise, LL Descriptor P+k should not be present. At least one LL Descriptor should be present. DscType P+k shall be set according to Table 2.56.

DscType Value	Meaning
0b00	LL Descriptor P+k not present; no blocks from the payload in the Payload Zone
0b01	LL Descriptor P+k is 2 octets; only Payload Length and LLID fields present; the payload has only one block (Nb = 0), which is being transmitted
0b10	LL Descriptor P+k is 5 octets; all fields present; TxBlockMap is 8 bits
0b11	LL Descriptor P+k is 6 octets; all fields present; TxBlockMap is 16 bits

Table 2.56: DscType values

Note: If a payload is not yet available to be transmitted, then DscType will be set to 0b00; see Section 2.

Note: An Empty Payload is still a payload (with a single block) and so requires an LL Descriptor.

The LL Descriptor field contains information about a specific payload and shall have the format shown in Figure 2.105.

LL Descriptor						
Payload Length (13 bits)	LLID (2 bits)	RFU (1 bit)	Nb (4 bits)	BlockSize (9 bits)	RFU (3 bits)	TxBlockMap (8 or 16 bits)

Figure 2.105: LL Descriptor field contents



Link Layer Specification

The Payload Length field shall include the total length of the payload, not just those blocks being transmitted, in octets. Therefore, Payload Length remains the same irrespective of which blocks are transmitted or retransmitted in any given packet. The value shall include any MIC but exclude any CRCs. For an Empty Payload, the value shall be 0.

The LLID field is described in Section 2.4 for Data Physical Channel PDUs, Section 2.6.1 for CIS PDUs, and Section 2.6.2 for BIS PDUs.

The Nb field is described in Section 2.7.4. If DscType is 0b10 or 0b11, Nb shall not be zero.

The BlockSize field shall specify the length of each block in the payload except the last one (see Section 2.7.4). If Nb is zero, then BlockSize shall be ignored.

The TxBlockMap field shall indicate which blocks of the payload are included in the Payload Zone. Block b of the payload (counting from 0 to Nb) shall be included if and only if bit b of TxBlockMap is set to 1; bits of TxBlockMap greater than Nb (if any) shall be set to 0. At least one bit should be set to 1.

If the TxBlockMap field has only one octet, the omitted bits 8 to 15 are treated as being set to 0. For example, a single-octet TxBlockMap field with the value 0x42 has the same meaning as a two-octet TxBlockMap field with the value 0x0042.

If the Tx Portion is not present in a format 1 packet, then the Payload Zone shall be empty, meaning no payloads can be transmitted in that packet.

If a DscType field corresponding to payload number n equals 0b00, then the packet shall only contain blocks from payloads with payload number greater than n if the Link Layer knows that the payload with payload number n contains at least one octet. A DscType field corresponding to an Empty Payload should not equal 0b00. This paragraph only applies to ACL connections. [ID28001_VSr03]

2.7.2.4 Extended Header

[HDT-PACK_VSr03]

The Extended Header, when it is present, can contain various additional pieces of information (called features) used by the Controller. All features are optional to include in a packet unless otherwise stated in this section.

The Extended Header consists of one or more extended header flags octets followed by zero or more extended header feature data octets, as shown in Figure 2.106.



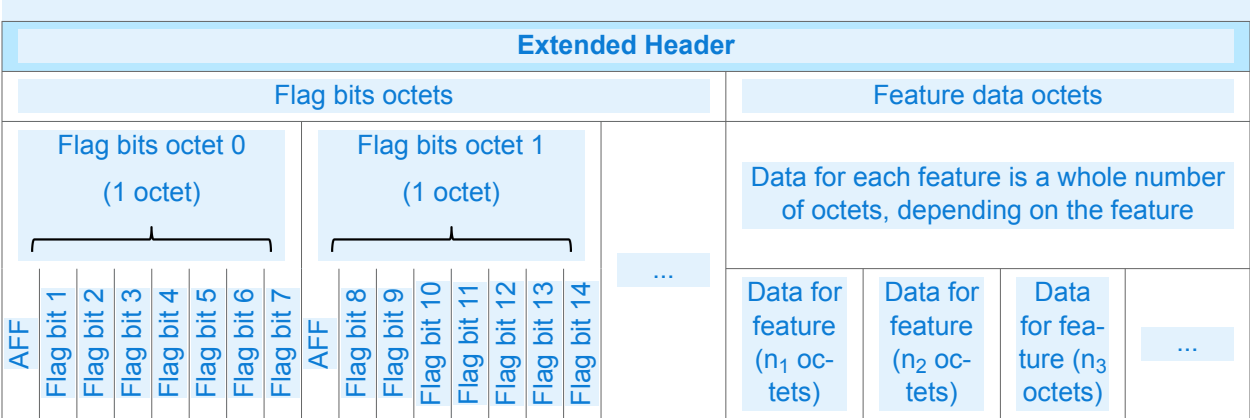


Figure 2.106: Extended Header – all fields optional except flags octet 0

Each extended header feature consists of a flag bit and zero or more data octets. If the flag bit is set to 1, then the feature is present and the Extended Header shall include any data octets specified for that feature. If the flag bit is set to 0, then the feature is absent and no data for that feature is included in the Extended Header.

Bit 0 of each extended header flags octet is the Additional Flags Follow (AFF) bit. The AFF bit shall have the value 0 if this is the last extended header flags octet and the value 1 if at least one more extended header flags octet follows. The remaining bits of each octet are flag bits, with the flag bit for feature number N being bit number B of octet O (O=0 is the first octet of the Extended Header), where $N = 7 \times O + B$. If there are F extended header flags octets, then all features with number greater than $7 \times F$ are absent. The last octet (i.e., with AFF=0) should have at least one flag bit set to 1.

The extended header feature data octets shall contain the data for those features that are present, in numerical order of feature number.

Table 2.57 lists the extended header features and their properties for packets sent on an ACL, Table 2.58 for packets sent on a CIS, and Table 2.59 for packets sent on a BIG. In each table, all feature numbers not listed are reserved for future use.

Feature Number	Feature	Data Octets	Meaning if Present	M/O
1	Extended Header feature data length	1	The total length of the extended header feature data octets. Shall be included in all format 0 packets that contain an Extended Header.	M
2	Link Quality	2	Link Quality (see Section 2.7.2.4.1)	C.1
3	CTE	1	Packet has a CTE; the data octet has the same format as the CTEInfo field defined in Section 2.5.2.	C.2

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Feature Number	Feature	Data Octets	Meaning if Present	M/O
4	Stop	0	Flow Control set to STOP (see Section 2.7.2.4.2).	M
5	NPI	0	The PDU is a Null PDU rather than an Empty PDU; this feature shall only be used in packet format 0 with an empty Payload Zone.	M
C.1: Mandatory if the Link Quality Feedback feature (Section 4.6.59) is supported, otherwise excluded.				
C.2: Mandatory if CTEs on the LE HDT PHY (Section 4.6.60) and Transmitting Constant Tone Extensions (Section 2.5.3) are supported, otherwise excluded. [ID28640_VSr03]				

Table 2.57: Extended header features for packets sent on an ACL

Feature Number	Feature	Data Octets	Meaning if Present	M/O
1	Extended Header feature data length	1	The total length of the extended header feature data octets. Shall be included in all format 0 packets that contain an Extended Header.	M
2	Link Quality	2	Link Quality (see Section 2.7.2.4.1)	C.1
3	NPI	0	The PDU is a Null PDU rather than an Empty PDU; this feature shall only be used in packet format 0 with an empty Payload Zone.	M
C.1: Mandatory if the Link Quality Feedback feature (see Section 4.6.59) is supported, otherwise excluded.				

Table 2.58: Extended header features for packets sent on a CIS

Feature Number	Feature	Data Octets	Meaning if Present	M/O
1	Extended Header feature data length	1	The total length of the extended header feature data octets. Shall be included in all format 0 packets that contain an extended header.	M

Table 2.59: Extended header features for packets sent on a BIG

For packets on ACL or CIS connections, the Link Layer shall not include a feature in the Extended Header unless it knows the peer device supports that feature.

The receiving Link Layer shall be able to interpret the structure of any Extended Header that contains only extended header features listed in this section; i.e., it shall be able to associate each data octet with the correct feature whether or not it supports that feature. The Link Layer shall be able to transmit and to interpret on reception any extended header feature where the M/O column of Table 2.57, Table 2.58, or Table 2.59 indicates that the feature is mandatory to support.



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Note: The previous paragraph does not require the Link Layer to transmit a feature in the Extended Header just because it is mandatory to support.

If the Link Layer receives an Extended Header where the flag bit is set for a feature that is listed in this section but it does not support, then it shall ignore the data for that feature. If the Link Layer receives an Extended Header where the flag bit is set for a feature number not listed in this section, then:

- If the packet used packet format 1 or included feature 1 (Extended Header feature data length), then the recipient shall ignore that flag bit and all higher-numbered ones.
- If the packet used packet format 0 and did not include feature 1, then the Link Layer shall consider the connection lost. If the Link Layer reports this event, then it shall use the error code *Invalid Packet Format* (0xTBD).

Note: In the second case, it is not possible to determine where the Extended Header ends and the Payload Zone begins.

2.7.2.4.1 Link Quality

[HDT-PACK_VSr03]

The Link Quality feature provides information about a packet (called the described packet) that was transmitted to the local device. The local device transmits this information in the Link Quality feature within the Extended Header of a packet.

The described packet shall be the most recent packet that the local device received from the remote device, whether or not all of the packet is valid (see Section 3.1).

This feature shall only be included in an Extended Header transmitted in a given connection event or CIS event if:

- The described packet is the most recent packet that the local device received in the same connection event or CIS event; or
- The local device is the Central, the packet containing the Link Quality information is the first packet that the Central transmits in that connection event or CIS event, the Peripheral transmitted the described packet in the most recent previous event where the Central transmitted a packet, and the described packet was transmitted after the last packet transmitted by the Central in that event.

The feature data octets shall have the format shown in Figure 2.107.



Link Quality		
RSSI (8 bits)	SNR (5 bits)	Preferred rate (3 bits)

Figure 2.107: Link Quality feature data octets format

The RSSI field indicates the received signal strength of the described packet in dBm. The value has type sint_8 and shall be either between -128 and +20 or be +127 to indicate that no RSSI is available. The reported value shall be within ±6 dB of the actual value.

The SNR field indicates the signal-to-noise ratio of the described packet in dB. The value shall be in the range 0 to 30 to indicate a specific value or 31 to indicate that no SNR is available. Values less than 0 dB or greater than 30 dB shall be reported as 0 or 30 respectively. The reported value shall be within ±3 dB of the actual value.

The Preferred rate field indicates that the device sending the information recommends that the receiving device uses a particular rate when transmitting packets other than those using short format. The rate shall be represented as specified in Table 2.54, except that 0b000 indicates that no recommendation is being made.

At least one field in the link quality information shall contain a specific value (as opposed to “not available” or “no recommendation”).

The receiving device may ignore some or all of the link quality data. Any use of this data by the receiving device is vendor-specific.

2.7.2.4.2 Flow control

[HDT-PACK_VSr03]

The Link Layer shall have a flow control state for each ACL connection it is part of. This state (but not the flow control state of the peer device) affects when the Link Layer can transmit payloads. The state has two values: STOP and GO.

When the flow control state of a connection is STOP, the Link Layer shall only transmit blocks on that connection from payloads that have had at least one block already acknowledged or that have an LLID field equal to 0b11. If the peer Link Layer receives any other valid block from the local Link Layer, it shall either process it normally or shall ignore the block. This does not affect any other part of a packet.

When the flow control state of a connection is GO, the flow control state does not restrict which payloads the Link Layer may transmit.

The initial value when a connection is created shall be GO. The value when the connection is on a GFSK PHY shall be GO.

When the Link Layer receives a packet with a valid PDU Header, it shall immediately set the flow control to STOP if the packet contains the Stop extended header feature and to GO otherwise. When the Link Layer receives a short format packet, it shall immediately set the flow control to GO. If the Link Layer receives a packet with an invalid PDU Header, then it shall not change the flow control state.

2.7.2.5 HEC-P Portion

[HDT-PACK_VSr03]

The HEC-P Portion shall contain a CRC of the other portions of the PDU Header Zone, generated as specified in Section 3.1.1.

The HEC-P Portion is only present in packet format 1.

2.7.3 Payload Zone

[HDT-PACK_VSr03]

The Payload Zone shall contain the payload blocks being transmitted in the packet. Only whole blocks shall be transmitted. In packet format 0, the only block (which can have zero length, producing an Empty PDU) shall be followed by a CRC of the PDU Header Zone and block contents. In packet format 1, each payload block shall be immediately followed by a CRC of its contents, except that both the block and CRC are omitted for an Empty Payload. The payloads shall be in numerical order of *PayloadNumber* and, within each payload, in numerical order of block number. The CRCs shall be generated as specified in Section 3.1.2.

For example, Figure 2.108 shows the format of a Payload Zone containing block 0 of payload cPN+0 and blocks 1, 2, and 4 of payload cPN+2 (where cPN is the value of *PayloadNumber* derived from the SN field of the packet containing the Payload Zone).

Payload Zone							
Payload cPN+0 block 0	CRC for cPN+0 block 0	Payload cPN+2 block 1	CRC for cPN+2 block 1	Payload cPN+2 block 2	CRC for cPN+2 block 2	Payload cPN+2 block 4	CRC for cPN+2 block 4

Figure 2.108: Example Payload Zone structure for packet format 1

In packet format 1, if the Payload Zone contains between 1 and 15 octets, then it shall be padded out to 16 octets using all-zero octets after the last CRC. The length information in the PDU Header Zone does not count these padding octets.

If the payload is encrypted, then the CRC is calculated on the encrypted data (including MIC); see Section 3.

2.7.4 Payload block structure

[HDT-PACK_VSr03]

Each payload shall be divided into between 1 and 16 blocks. The number of blocks in a payload is indicated by the Nb field for that payload: the value of the Nb field shall be one less than the number of blocks that the payload is divided into (e.g., Nb=0 means one block, Nb=1 means two blocks, and Nb=15 means 16 blocks).

Each block except the last one shall have the same length, which shall be between 32 and 511 octets (this is the value of the BlockSize field of the LL Descriptor). The length of the last block of a payload with more than one block shall be between 32 and BlockSize + 32 octets. The length of a payload with only one block shall be between zero and 1023 octets. Different payloads with the same Payload Length may have different BlockSize values.

Note: The length of the last block is always Payload Length – Nb × BlockSize octets.

Note: A payload can only be sent using packet format 0 if Nb=0 and the payload length is not greater than 510 octets.

Note: A payload of less than 32 octets always has Nb=0.

If the payload is encrypted, then it is split into blocks after it has been encrypted and the MIC appended. A received payload can only be completely decrypted and the MIC checked after all the blocks of the payload have been correctly received and reassembled, though individual blocks can be decrypted as soon as they are received.



3 BIT STREAM PROCESSING

Bluetooth devices shall use the bit stream processing schemes as defined in the following sections.

Figure 3.1 shows the bit stream processing for PDUs on the LE Uncoded PHYs.

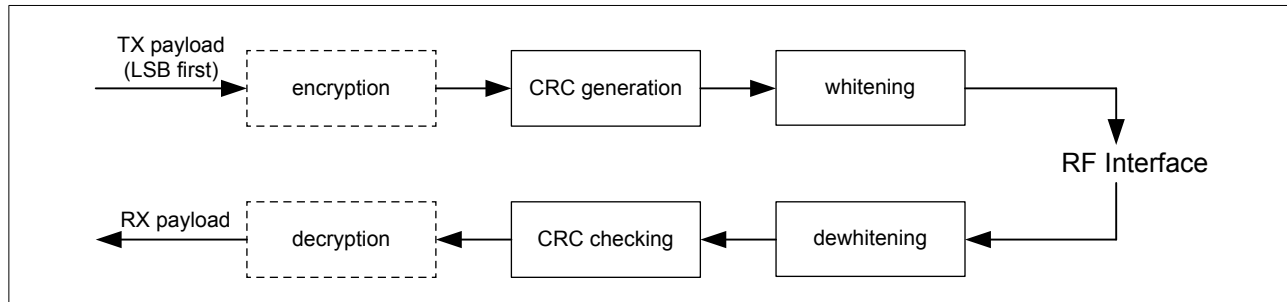


Figure 3.1: Payload bit processes for the LE Uncoded PHYs

Figure 3.2 shows the bit stream processing for PDUs on the LE Coded PHYs.

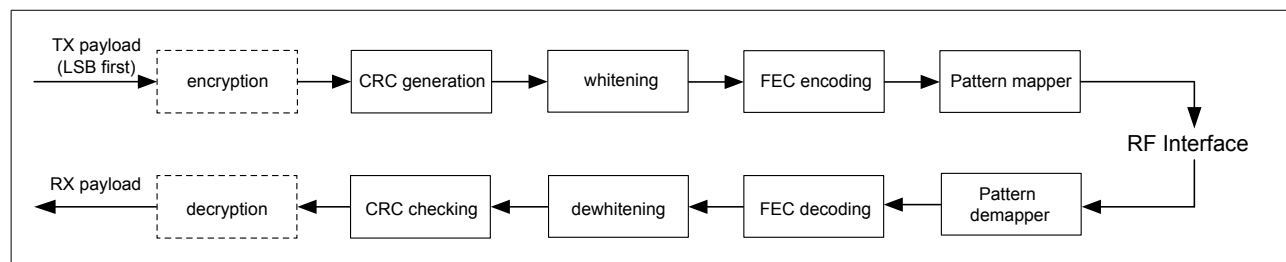


Figure 3.2: Bit stream processing for the LE Coded PHYs [ID28364_VSr02]

Figure 3.3 shows the bit stream processing for the Control Header, PDU Header, and PDUs on the LE HDT PHY. [HDT-PHY_VSr01]

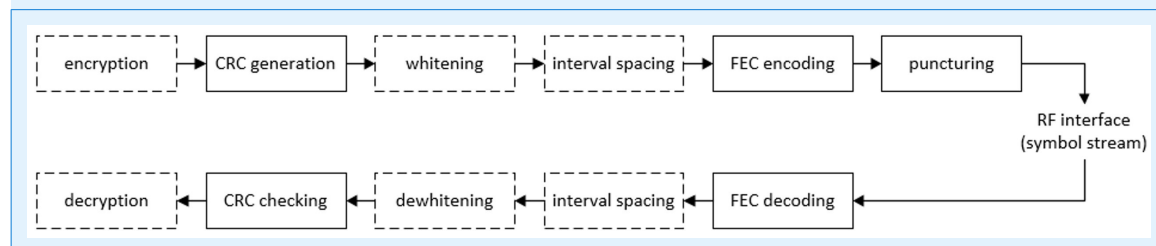


Figure 3.3: Bit stream processing for LE HDT PDUs [HDT-PHY_VSr01] [ID27963_VSr02]

The top row of Figure 3.3 shows the bit stream processing at the transmitter. After puncturing, the processed bitstream is available to the RF interface. The bottom row of Figure 3.3 shows the bit stream processing at the receiver, where the RF interface



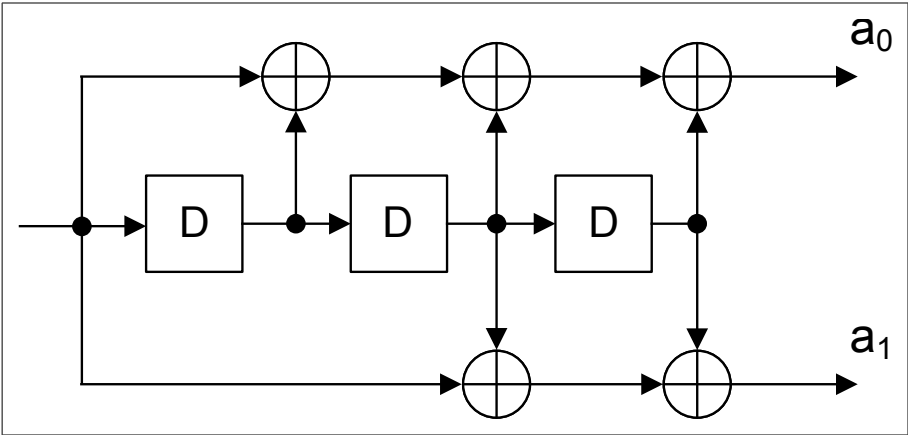


Figure 3.8: Convolutional Forward Error Correction encoder

3.3.2 Pattern mapper

The pattern mapper converts each bit from the convolutional FEC encoder into P symbols, where the value of P depends on the coding scheme in use, according to Table 3.1 (the output sequences are in transmission order):

Input bit from the convolutional FEC encoder	Output sequence when P=1 (used by S=2)	Output sequence when P=4 (used by S=8)
0	0	0011
1	1	1100

Table 3.1: Pattern mapper inputs and outputs

3.4 Bit stream processing for the LE HDT PHY

[HDT-PHY_VSr01]
[ID27963_VSr02]

This section only applies to the LE HDT PHY. Each sequence of bits shall go through the following processes, in order:

- Encryption
- CRC generation
- Whitening
- Interval spacing (only for the Payload Zone in packet format 1)
- FEC encoder
- Puncturing

In the short packet format, there is only one sequence of bits: the Control Header bit sequence. In packet format 0, there are two sequences of bits: the Control Header bit

sequence and the PDU Header and Payload bit sequence. In packet format 1, there are at least two sequences of bits: The Control Header bit sequence and the PDU Header bit sequence. If there is a payload, then a Payload bit sequence is also present.

The PDU Header and Payload use whitening. The Control Header is not whitened.

In packet format 1, the Payload bit sequence is split into PHY Intervals as described in Section 3.4.2.

All sequences use FEC encoding at a coding rate of 1/2.

Puncturing may be used to increase the coding rate to 2/3, 3/4, or 15/16. The effective bit rate of the Control Header is 2 Mb/s and its coding rate is 1/2. [ID28319_VSr03]

The coding rates used in the PDU Header and Payload in packet format 0 and in the Payload Zone in packet format 1 are shown in Table 3.2.

Name	HDT2	HDT3	HDT4	HDT6	HDT7.5
Effective Bit Rate	2 Mb/s	3 Mb/s	4 Mb/s	6 Mb/s	7.5 Mb/s
Rate Indicator in Control Header	0b001	0b010	0b011	0b100	0b101
Coding Rate	1/2	3/4	2/3	3/4	15/16

Table 3.2: Payload rate indicator and coding [ID28319_VSr03]

The coding rates used in the PDU Header in packet format 1 are shown in Table 3.3.

Name	HDT2	HDT3	HDT4	HDT6	HDT7.5
Effective Bit Rate	2 Mb/s	2 Mb/s	3 Mb/s	4 Mb/s	4 Mb/s
Rate Indicator in Control Header	0b001	0b010	0b011	0b100	0b101
Coding Rate	1/2	1/2	1/2	1/2	1/2

Table 3.3: Packet format 1 PDU Header rate indicator and coding [ID28319_VSr03]

3.4.1 Data whitening

[HDT-PHY_VSr01]

Data whitening is carried out by XORing the output bits with a pseudo-random sequence. Data whitening is used to avoid long sequences of zeros or ones (e.g., 0b0000000 or 0b1111111) in the data bit stream.

Whitening shall be applied on the PDU Header bits (including HEC-P bits if present) and Payload bits (including 32-bit CRCs).

Whitening shall not be applied on the Control Header, including HEC-C and termination bit sequence.

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The whitener and de-whitener are defined in the same way. The whitener and de-whitener use a 15-bit linear feedback shift register with polynomial $x^{15} + x + 1$. Its structure is shown in Figure 3.9. The shift register is initialized with a sequence that is derived from the physical channel index in which the packet is transmitted in the following manner:

- Position 0 is set to 1.
- Positions 1 to 4 are RFU.
- Positions 5 to 12 are set to the (8-bit) channel index (see Section 1.4.1) of the channel used when transmitting or receiving, from the most significant bit in position 5 to the least significant bit in position 12.
- Position 13 is set to 1.
- Position 14 is set to 0.

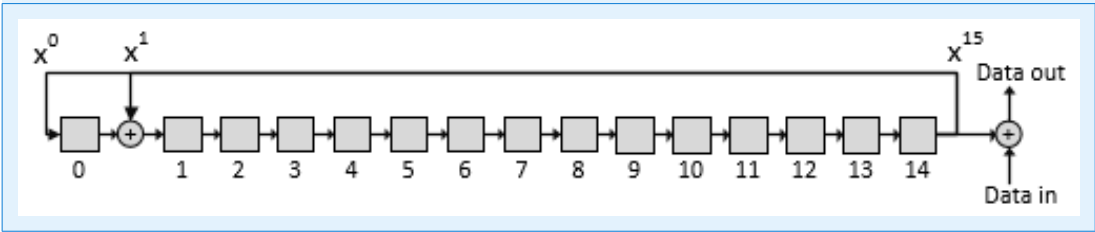


Figure 3.9: The LFSR used to generate whitening on the LE HDT PHY [ID27963_VSr02]

Whitening starts at the first PDU Header bit after the Control Header and ends at the last data bit of the Payload Zone. In packet format 1, after the whitening process is paused at the termination sequence bits of the PDU Header or a PHY Interval, whitening continues at the first data bit of the next PHY Interval by resuming at the same state it was paused.

3.4.2 Interval spacing

[HDT-PHY_VSr01]

In packet format 1 only, the Payload Zone shall be divided into “intervals”, each of which consists of the number of octets specified by the PHY INT field of the Control Header. This process of dividing the Payload Zone into intervals is referred to as interval spacing. The number of octets for each PHY INT value is shown in Table 3.4.

PHY INT	HDT2	HDT3	HDT4	HDT6	HDT7.5
0b00	16	24	32	48	60
0b01	32	48	64	96	120



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PHY INT	HDT2	HDT3	HDT4	HDT6	HDT7.5
0b10	48	72	96	144	180
0b11	64	96	128	192	240

Table 3.4: Number of data octets per PHY Interval

The number of octets in Table 3.4 include the CRC bits but do not include the termination sequence bits and padding bits for symbol completion.

Note: The number of octets is derived from the number of symbols in the interval as specified in [Vol 6] Part A, Section 7.1.

If the number of octets in the Payload Zone is not a multiple of the number of data octets per PHY Interval, then the last PHY Interval is shorter than the other PHY intervals.

3.4.3 Convolutional encoder

[HDT-PHY_VSr01]

The convolutional FEC encoder uses a non-systematic, non-recursive rate 1/2 code with a constraint length of 6 (32 internal states). Its generator polynomials are:

$$G_0(x) = 1 + x^2 + x^4 + x^5$$
$$G_1(x) = 1 + x + x^2 + x^3 + x^5$$

The initial state of the convolutional FEC encoder is all zeros. An input sequence of five consecutive zeros shall be appended to the end of the input data to bring the convolutional FEC encoder back to its initial state. This sequence is known as the termination sequence bits.

Each sequence of bits contains its own termination sequence. In packet format 1, a termination sequence is inserted at the end of all PHY Interval bit streams after the last PHY Interval data octet.

Figure 3.10 specifies the encoder operation. Squares represent bit storage operations; circles represent XOR. Bit a_0 is transmitted first.



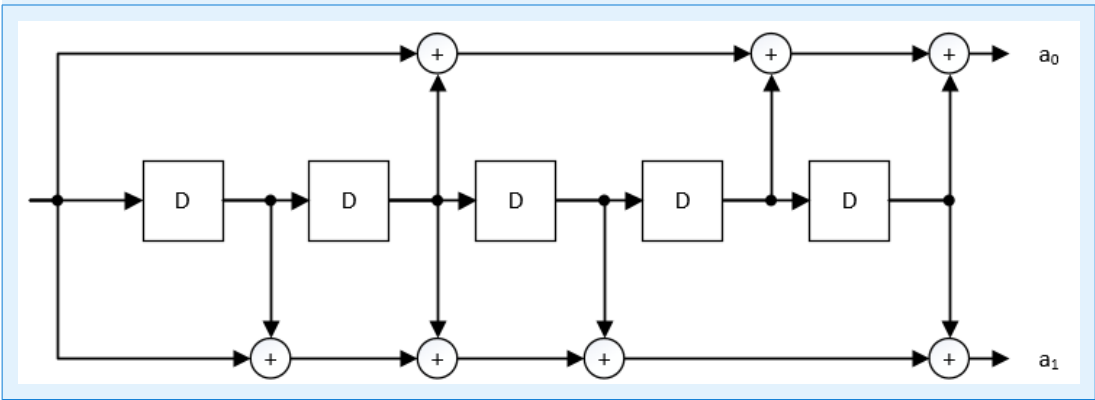


Figure 3.10: Forward Error Correction Encoder for the LE HDT PHY [ID27963_VSr02]

3.4.4 Puncturing

[HDT-PHY_VSr01]

When puncturing is used, some of the bits from the output of the FEC encoder are omitted. The puncturing patterns used are detailed in Table 3.5, where a 1 represents a bit that is retained and a 0 represents a bit that is omitted. The pattern is repeated until the end of the encoded sequence is reached. The last puncturing pattern repetition is truncated where necessary to match the data length.

Coding Rate	Puncturing Pattern
1/2	[1 1]
2/3	[1 1 0 1]
3/4	[1 1 0 1 0 1]
15/16	[1 1 0 1 1 0 1 0 1 0 0 1 0 1 0 1 1 0 1 0 0 1 0 1 0 1 1 0 0 1]

Table 3.5: Puncturing patterns for the LE HDT PHY [ID27963_VSr02]

Using the 2/3 pattern as an example, [1 1 0 1] means that, of each group of 4 bits output by the FEC encoder, the first, second, and fourth are retained but the third bit is omitted.

Note: The puncturing pattern [1 1] is equivalent to no puncturing.

